BUSINESS INTELLIGENCE / C1

Who contributes the recorded purchase value?

Customer concentration in UCI Online Retail



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29 September 2026

The Online Retail case

541.909

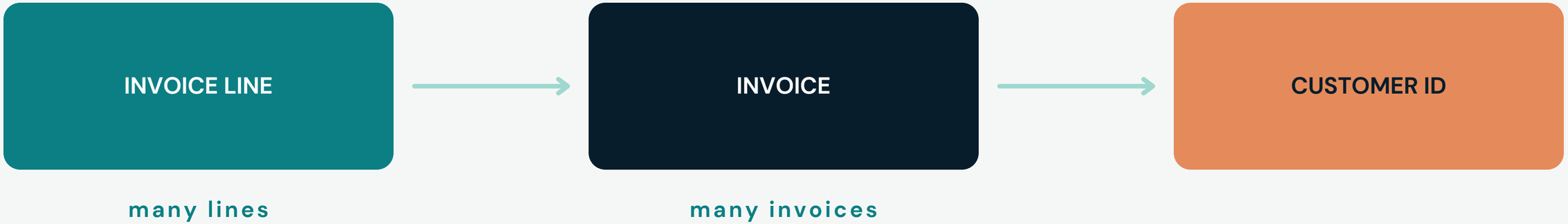
rows in the public file · 516.384 used in our study

UCI describes a UK non-store gift retailer with many wholesale customers. We study all invoice lines from Dec 2010 to Nov 2011. We leave out the final nine days of Dec 2011 so each month in the case is complete.

COLUMN	WHAT IT MEANS
InvoiceNo	Invoice number. A C prefix flags a cancellation record.
StockCode	Product code
Description	Product name
Quantity	Signed number of units
InvoiceDate	Invoice date and time
UnitPrice	Price of one unit (£)
CustomerID	Customer number (sometimes blank)
Country	Country where the customer lives

SOURCE UCI Machine Learning Repository · Chen (2015) · CC BY 4.0

What the file records



An invoice line is one item entry on an invoice. It records the product, quantity and unit price. One invoice can contain several lines, and several invoices can belong to the same customer.

A small number of customers record much higher purchase value

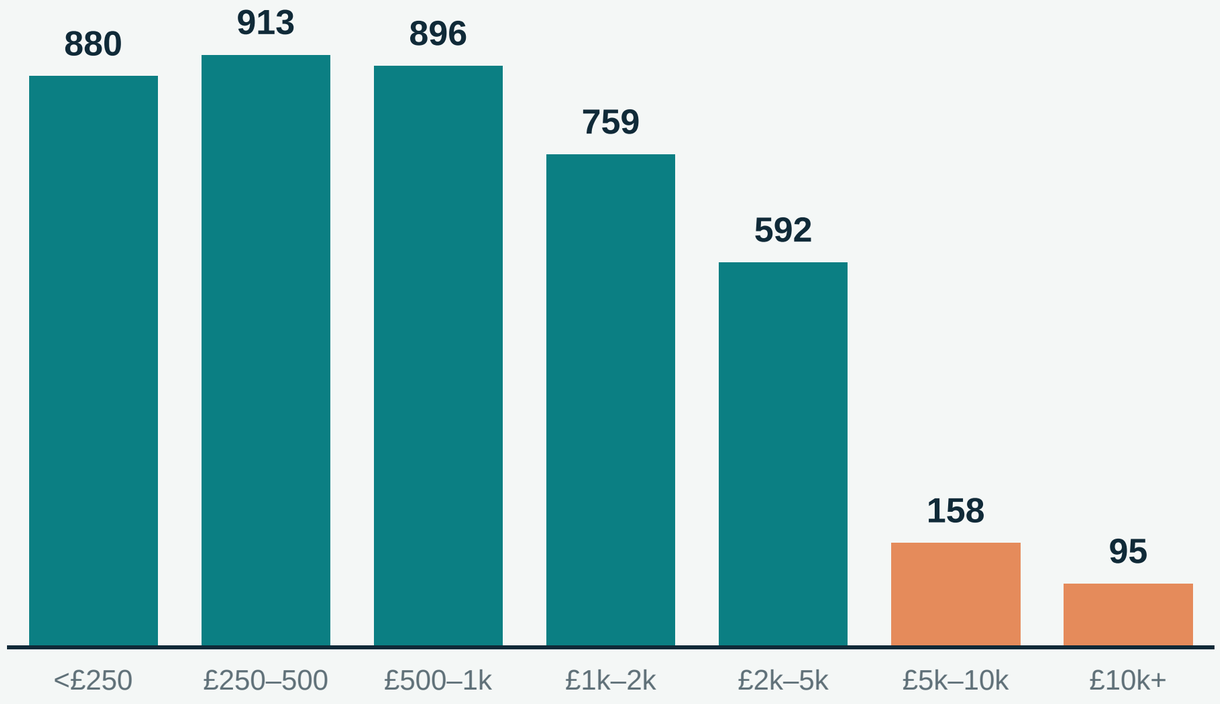
4,293

customers with an ID

Median £652.80

Mean £1,917.78

63% of identified customers recorded less than £1,000, while only 95 customers recorded £10,000 or more. This wide difference motivated our next question: what share of known customer value comes from the top 10%?



Customers by recorded value per ID (£)

Cancellation invoices vary across months

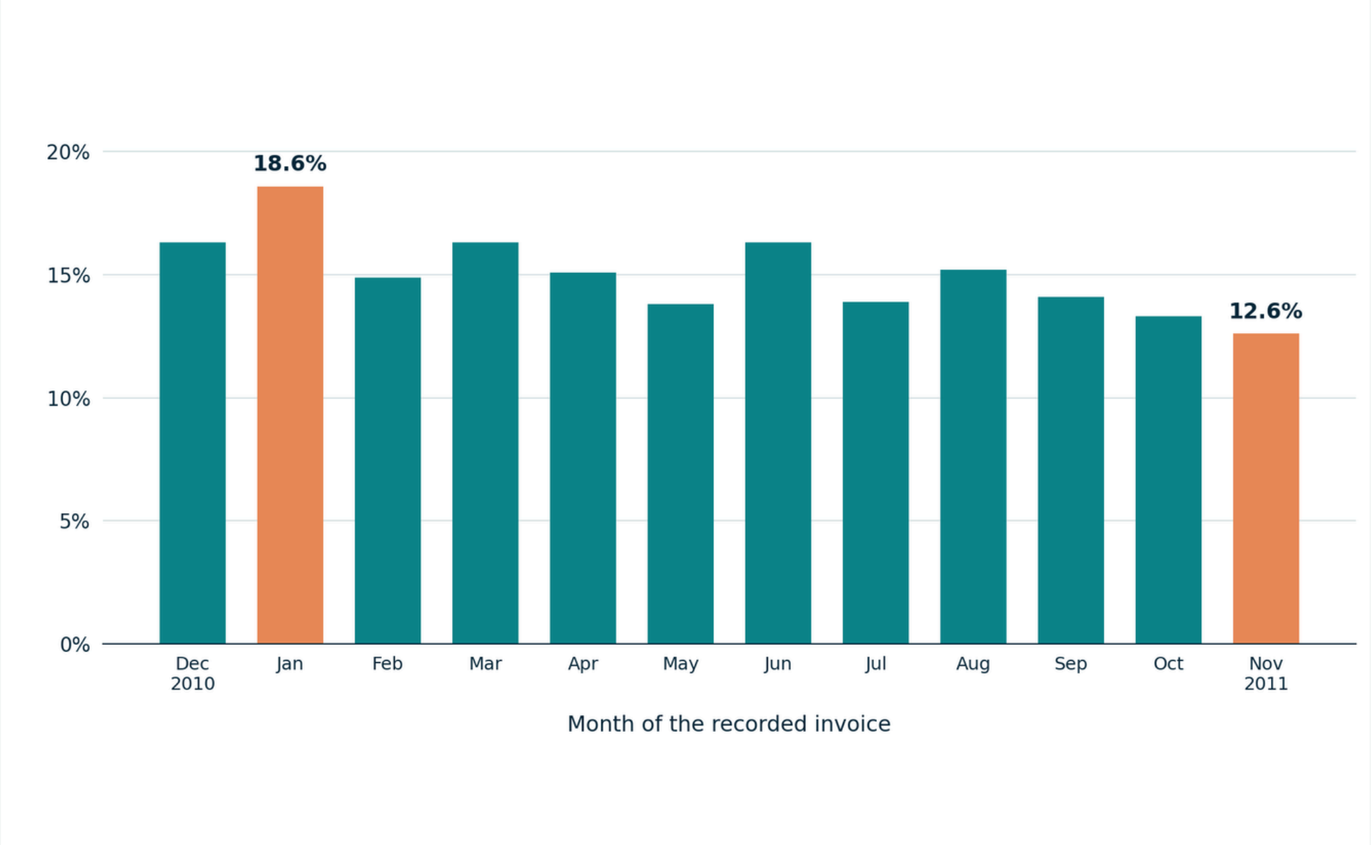
3.283

eligible cancellation invoices

High: 18.6% (Jan)

Low: 12.6% (Nov)

This difference motivated Candidate B's question: in which months do cancellation invoices make up a larger share, and where should an operations team look more closely?



From W1 to C1: why customer concentration?

Candidate A

How concentrated is recorded purchase value?

A small number of wholesale accounts may contribute a large share of purchases. Measuring that share would help a commercial team identify which accounts deserve closer review and monitor how dependent the recorded portfolio is on them.

Candidate B

When and where do cancellation invoices appear?

The dataset identifies invoices recorded as cancellations. Comparing their frequency by month and country could point to areas for an operations review. We cannot link each cancellation to its original purchase or see its reason, so we cannot calculate a purchase-level cancellation rate

We chose A because its customer-value denominator and ID coverage can be measured and tested. B remains useful for monitoring, but lacks the purchase links needed for a cancellation rate.

What we will measure, and why

K10

The percentage of purchase value from the highest-value 10% of customers with an ID.

Shows how much the business depends on its largest identifiable customers.

HOW WE MEASURE IT

Value from the top 10% of known customers ÷ value from all known customers × 100

IDC

The percentage of recorded purchase value that has a CustomerID.

Shows how much purchase value the customer ranking leaves out.

HOW WE MEASURE IT

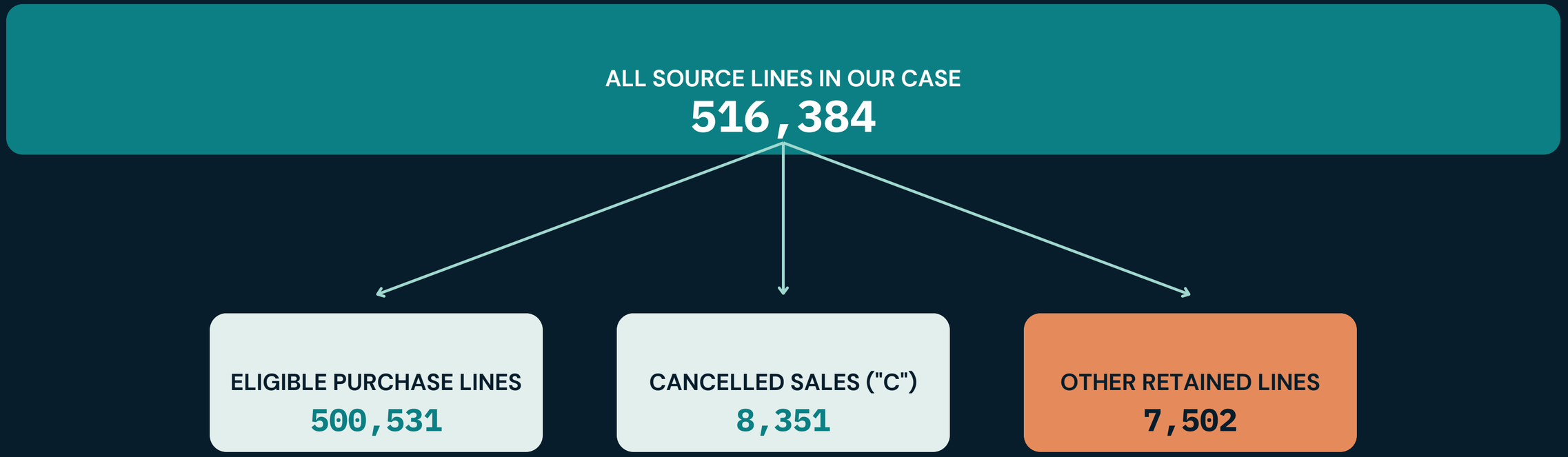
Eligible purchase value with a CustomerID ÷ all eligible purchase value × 100

Data quality flags and our treatment

FLAG	LINES	DECISION
Missing CustomerID	127,216	Keep in eligible value; omit only from customer ranking
Exact repeated rows	4,989	Keep, there is no unique line ID to prove they are accidental duplicates.
Outside merchandise code rule	5,072	Exclude, these codes may represent charges or other non-product entries.
UnitPrice ≤ 0	2,463	Exclude, a zero or negative price does not represent a positive recorded charge and we are evaluating revenue.
Description missing	1,439	Keep, the description is not needed.

We flag these rows rather than delete them from the source.

Every source line remains auditable



The Five Cs on this case

CLEAN

We checked missing IDs, prices, product codes and repeated rows. We documented which records enter each measure.

CONSISTENT

We use the same purchase and KPI rules across the 12 complete months

CONFORMED

We use one source and make no external join, so cross-source agreement does not apply here.

CURRENT

The records are from 2010–2011. They support a historical case study, not a decision about today’s customers.

COMPREHENSIVE

Some purchases have no CustomerID. The file also lacks costs and links from cancellations to original purchases.

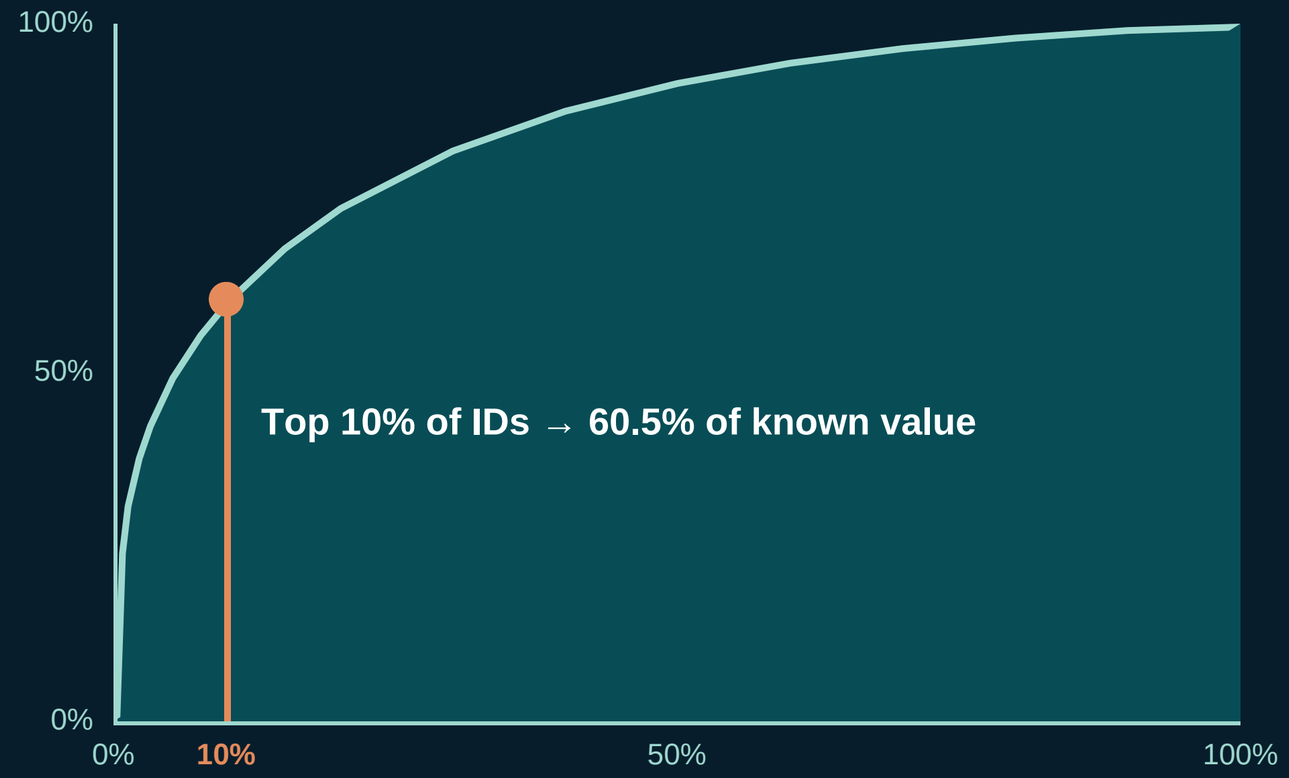
The top 10% of known customers account for most known purchase value

K10 60.5%

from the top 10% of IDs (430)

IDC 85.7%

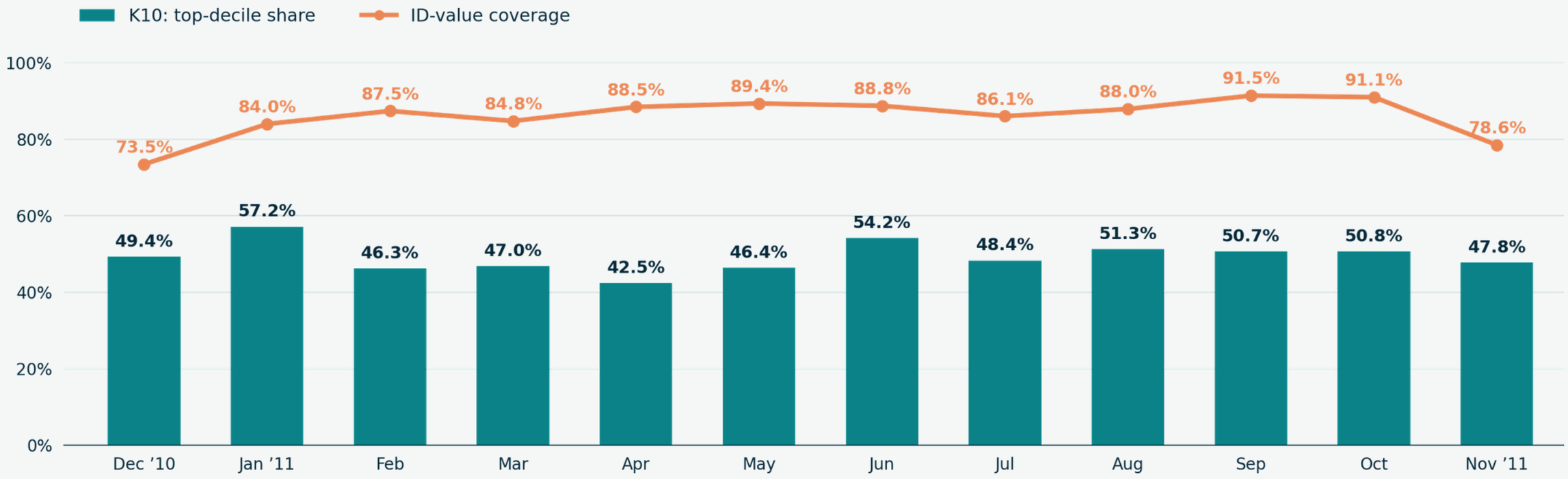
of eligible value linked to an ID



K10 (60.5%) shows that the top 10% of identified customers account for most known purchase value. IDC (85.7%) means 14.3% of eligible value cannot enter that ranking. We must review major accounts and ID capture together.

KPIs change the monthly reading

Both series use a 0–100% scale. K10 re-ranks known customer IDs within each month.



October to November: K10 fell from 50.8% to 47.8%, while ID coverage fell from 91.1% to 78.6%. Check missing IDs before interpreting the K10 change.

The concentration finding changes little under alternative data rules

ROW RULE	TOP-10% SHARE
Baseline merchandise rule (4,293 IDs)	60.50%
Drop exact repeated rows	60.58%
Strict five-digit codes only	60.41%
Remove the largest known ID	59.22%

K10 remains between 59.2% and 60.6% when we change the duplicate and product-code rules or remove the largest customer. These checks support the finding for known IDs, but cannot recover purchases without a CustomerID.

What the evidence supports

EVIDENCE

The top 10% of identified customers account for 60.5% of known purchase value

K10 remains between 59.2% and 60.6% under alternative data rules. However, 14.3% of eligible value has no CustomerID.

PROPOSED ACTIONS

Review major accounts and improve ID capture together.

Inspect the invoices behind high-value IDs and investigate missing IDs.

Thank you